

Serie 27

BILINEAR FORMS

1. Consider the real matrix

$$A = \begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix} \in M_{3 \times 3}(\mathbb{R}).$$

Find a matrix $P \in O_3(\mathbb{R})$ such that $P^{-1}AP$ is diagonal.

2. *Example regarding special relativity.* Define the symmetric bilinear forms $s: \mathbb{R}^4 \rightarrow \mathbb{R}$ for all $v = (x, y, z, t)^T$ and $v' = (x', y', z', t')^T$ in $\mathbb{R}^4$ by

$$s(v, v') := xx' + yy' + zz' - ctt',$$

where $c > 0$ is a fixed parameter. The space $M := (\mathbb{R}^4, s)$ is called *Minkowski space* (sometimes Minkowski spacetime) and the parameter c is called *light speed*. We use the normalization $c = 1$.

A linear map $F: M \rightarrow M$ is called *isometry* or *Lorentz transformation*, if

$$\forall v, w \in \mathbb{R}^4: s(F(v), F(w)) = s(v, w).$$

- (a) Show that every isometry is bijective.
- (b) Show that the following endomorphisms are isometries of M :
 - i. Left multiplication with $\left(\begin{array}{c|c} T & 0 \\ \hline 0 & \pm 1 \end{array} \right)$ für jedes $T \in O(3)$.
 - ii. A *Lorentz boost* in x -direction with speed $v < c = 1$, given by left multiplication with the matrix

$$B := \begin{pmatrix} \gamma & & -v\gamma \\ & 1 & \\ -v\gamma & & \gamma \end{pmatrix}$$

for $\gamma := 1/\sqrt{1-v^2}$.

- (c) The subset $\{x \in M \mid s(x, x) = 0\}$ is called *light cone in M* . Prove the „relativistic football theorem“: Every linear isometry φ with $\det(\varphi) = 1$ has an eigenvector in the light cone.

Remark. For $c \rightarrow \infty$ the light cone approaches the subspace $\{t = 0\}$ and the statement reduces to the classical case.

3. Let n be a positive integer. Let V be the space of all $n \times n$ matrices over the field of complex numbers. For a matrix $A = (a_{i,j})_{i,j} \in V$ recall that the trace is given by $\text{tr}(A) = \sum_{i=1}^n a_{i,i}$.

(a) Show that the equation

$$f(A, B) = n\text{tr}(AB) - \text{tr}(A)\text{tr}(B)$$

defines a bilinear form f on V . Is it true that $f(A, B) = f(B, A)$ for all A, B ?

- (b) Let f be the bilinear form defined in part (3a). Show that f is degenerate (not non-degenerate). Let V_1 be the subspace of V consisting of the matrices of trace 0, and let f_1 be the restriction of f to V_1 . Show that f_1 is non-degenerate.
- (c) Let f be the bilinear form defined in part (3a), and let V_2 be the subspace of V consisting of all matrices A such that the trace of A is zero, and $A^* = -A$ (where A^* is the conjugate transpose of A). Denote by f_2 the restriction of f to V_2 . Show that f_2 is negative definite, i.e. that $f_2(A, A) < 0$ for each non-zero A in V_2 .

4. Let

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \in \text{SO}(3),$$

and assume that $A \neq I$.

- (a) Show that 1 is an eigenvalue of A .
- (b) Determine explicitly a vector generating the axis of rotation of A .
- (c) Let $\theta \in [0, \pi]$ be the angle of rotation associated with A . Show that

$$\cos \theta = \frac{\text{tr}(A) - 1}{2}.$$

5. Let

$$A \in \text{SO}(3),$$

that is,

$$A^T A = I, \quad \det(A) = 1.$$

- (a) Prove that

$$-1 \leq \text{tr}(A) \leq 3.$$

Hint: Use Exercise 4.

- (b) Show that

$$\text{tr}(A) = 3 \iff A = I.$$

(c) Show that

$$\operatorname{tr}(A) = -1$$

if and only if A is a rotation of angle π .

6. Let $P_1, P_2 \subset \mathbb{R}^3$ be two distinct planes through the origin. For $i = 1, 2$, let

$$\sigma_i : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

denote the reflection in the plane P_i .

Assume that the angle between the planes is $\theta \in (0, \pi/2]$, meaning that the angle between their normal vectors is θ .

(a) Show that the composition

$$T = \sigma_2 \circ \sigma_1$$

is an orientation-preserving orthogonal transformation.

(b) Show that the line

$$L = P_1 \cap P_2$$

is pointwise fixed by T .

(c) Prove that T is a rotation around the axis L and determine its angle of rotation explicitly in terms of θ .